DBMS Tables Normalization Lab

Question 1

Problem Statement:

Normalize the table up to 3NF. **Dataset:**

OrderID	CustomerName	Product	Price
1	Anuj	Laptop	50000
1	Anuj	Mouse	500
2	Amit	Laptop	50000
2	Amit	Keyboard	1500
3	Ravi	Mouse	500
4	Sita	Laptop	50000
5	Mohan	Keyboard	1500
6	Riya	Mouse	500
7	Karan	Laptop	50000
8	Neha	Keyboard	1500

Answer:

Orders Table:

Order ID	CustomerName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha

Products Table

Product	Price
Laptop	50000
Mouse	500
Keyboard	1500

Order Details Table

OrderID	Product
1	Laptop
1	Mouse
2	Laptop
2	Keyboard
3	Mouse
4	Laptop
5	Keyboard
6	Mouse
7	Laptop
8	Keyboard

Question 2

Problem Statement:

Identify dependencies and normalize to 3NF. **Dataset:**

EmpID	EmpName	DeptID	DeptName
101	Anuj	D1	HR
102	Amit	D2	IT
103	Ravi	D1	HR
104	Sita	D3	Finance
105	Mohan	D2	IT
106	Riya	D1	HR
107	Karan	D3	Finance
108	Neha	D2	IT
109	Pooja	D1	HR
110	Rahul	D3	Finance

Answer:

Employees Table:

EmpID	EmpName
101	Anuj
102	Amit
103	Ravi
104	Sita
105	Mohan
106	Riya
107	Karan
108	Neha
109	Pooja
110	Rahul

Employee-Department Table :-

EmpID	DeptID
101	D1
102	D2
103	D1
104	D3
105	D2
106	D1
107	D3
108	D2
109	D1
110	D3

Department table:

DeptID	DeptName
D1	HR
D2	IT
D3	Finance

Dependencies:

EmpID → EmpName, DeptID

DeptID → DeptName

EmpID → DeptName *(transitive)*

Question 3

Problem Statement:

Remove transitive dependency and normalize. **Dataset:**

StudentID	Subject	Teacher	TeacherPhone
1	Math	Ravi	9999
1	Science	Ravi	9999
2	English	Sita	8888
3	Math	Ravi	9999
4	Science	Ravi	9999
5	English	Sita	8888
6	Math	Ravi	9999
7	Science	Ravi	9999
8	English	Sita	8888
9	Math	Ravi	9999

Answer:

Students_subject Table:

StudentID	Subject
1	Math
1	Science
2	English
3	Math
4	Science
5	English
6	Math
7	Science
8	English
9	Math

Subject_teacher Table:

Subject	Teacher
Math	Ravi
Science	Ravi
English	Sita

Teachers Details table:

Teacher	TeacherPhone
Ravi	9999
Sita	8888

Question 4:

Problem Statement:
Normalize to 3NF and identify all dependencies. **Dataset:**

BookID	Author	Publisher	PublisherAddress
B1	A1	P1	Delhi
B2	A2	P2	Mumbai
B3	A1	P1	Delhi
B4	A3	P3	Pune
B5	A2	P2	Mumbai
B6	A1	P1	Delhi
B7	A3	P3	Pune
B8	A2	P2	Mumbai
B9	A1	P1	Delhi
B10	A3	P3	Pune

Answer:

Books Details:

BookID	Author
B1	A1
B2	A2
B3	A1
B4	A3
B5	A2
B6	A1
B7	A3
B8	A2
B9	A1
B10	A3

Book_Publishers:

BookID	Publisher
B1	P1
B2	P2
B3	P1
B4	P3
B5	P2
B6	P1
B7	P3
B8	P2
B9	P1
B10	P3

Publishers Table:

Publisher	PublisherAddress
P1	Delhi
P2	Mumbai
P3	Pune

Question 5:

Problem Statement

Find candidate key and normalize to 3NF. **Dataset:**

OrderID	ProductID	ProductName	CustomerID	CustomerName
1	P1	Laptop	C1	Anuj
1	P2	Mouse	C1	Anuj
2	P1	Laptop	C2	Amit
2	P3	Keyboard	C2	Amit
3	P2	Mouse	C3	Ravi
4	P1	Laptop	C4	Sita
5	P3	Keyboard	C5	Mohan
6	P2	Mouse	C6	Riya
7	P1	Laptop	C7	Karan
8	P3	Keyboard	C8	Neha

Answer:

Order Details:

OrderID	ProductID
1	P1
1	P2
2	P1
2	P3
3	P2
4	P1
5	P3
6	P2
7	P1
8	P3

Order_Customer Details

OrderID	CustomerID
1	C1
2	C2
3	C3
4	C4
5	C5
6	C6
7	C7
8	C8

Customer Detail:

CustomerID	CustomerName
C1	Anuj
C2	Amit
C3	Ravi
C4	Sita
C5	Mohan
C6	Riya
C7	Karan
C8	Neha

Product details:

ProductID	ProductName
P1	Laptop
P2	Mouse
P3	Keyboard

Candidate key is: (OrderID, ProductID)

Question 6

Problem Statement:

Normalize the student-course table up to 3NF.

StudentID	StudentName	Course	Instructor
1	Anuj	Math	Ravi
1	Anuj	Science	Ravi
2	Amit	English	Sita
3	Ravi	Math	Ravi
4	Sita	Science	Ravi
5	Mohan	English	Sita
6	Riya	Math	Ravi
7	Karan	Science	Ravi
8	Neha	English	Sita
9	Pooja	Math	Ravi

Answer:

Student Details:

StudentID	StudentName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha
9	Pooja

Course details:

StudentID	Course
1	Math
1	Science
2	English
3	Math
4	Science
5	English
6	Math
7	Science
8	English
9	Math

Instructor Details:

Course	Instructor
Math	Ravi
Science	Ravi
English	Sita

Question 7

Problem Statement:

Normalize employee-project data to 3NF. **Dataset:**

EmpID	ProjectID	ProjectName	Manager
101	P1	Alpha	Raj
101	P2	Beta	Raj
102	P1	Alpha	Raj
103	P3	Gamma	Simran
104	P2	Beta	Raj
105	P3	Gamma	Simran
106	P1	Alpha	Raj
107	P2	Beta	Raj
108	P3	Gamma	Simran
109	P1	Alpha	Raj

Answer:

Employee_Projects Table :-

EmpID	ProjectID
101	P1
101	P2
102	P1
103	P3
104	P2
105	P3
106	P1
107	P2
108	P3
109	P1

Projects Table:-

ProjectID	ProjectName	Manager
P1	Alpha	Raj
P2	Beta	Raj
P3	Gamma	Simran

Question 8

Problem Statement: Normalize sales data up to 3NF.

SaleID	Product	Category	Price
1	Laptop	Electronics	50000
2	Mouse	Electronics	500
3	Keyboard	Electronics	1500
4	Shirt	Clothing	1000
5	Jeans	Clothing	2000
6	Laptop	Electronics	50000
7	Mouse	Electronics	500
8	Shirt	Clothing	1000
9	Jeans	Clothing	2000
10	Keyboard	Electronics	1500

Answer:

Sales Details:-

SaleID	Product
1	Laptop
2	Mouse
3	Keyboard
4	Shirt
5	Jeans
6	Laptop
7	Mouse
8	Shirt
9	Jeans
10	Keyboard

Product Details:-

Product	Price
Laptop	50000
Mouse	500
Keyboard	1500
Shirt	1000
Jeans	2000

Category Details:-

Product	Category
Laptop	Electronics
Mouse	Electronics
Keyboard	Electronics
Shirt	Clothing
Jeans	Clothing

Question 9

Problem Statement:

Normalize hospital patient records to 3NF. **Dataset:**

PatientID	PatientName	Doctor	DoctorSpecialization
1	Anuj	Dr. Ravi	Cardiology
2	Amit	Dr. Sita	Neurology
3	Ravi	Dr. Ravi	Cardiology
4	Sita	Dr. Sita	Neurology
5	Mohan	Dr. Ravi	Cardiology
6	Riya	Dr. Sita	Neurology
7	Karan	Dr. Ravi	Cardiology
8	Neha	Dr. Sita	Neurology
9	Pooja	Dr. Ravi	Cardiology
10	Rahul	Dr. Sita	Neurology

Answer:

Patient Table:-

PatientID	PatientName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha
9	Pooja
10	Rahul

Patient-Doctor table:-

PatientID	Doctor
1	Dr. Ravi
2	Dr. Sita
3	Dr. Ravi
4	Dr. Sita
5	Dr. Ravi
6	Dr. Sita
7	Dr. Ravi
8	Dr. Sita
9	Dr. Ravi
10	Dr. Sita

Doctors Details:-

Doctor	DoctorSpecialization
Dr. Ravi	Cardiology
Dr. Sita	Neurology

Question 10

Problem Statement:

Normalize university data to 3NF. **Dataset:**

StudentID	Dept	HOD
1	CSE	Dr. A
2	IT	Dr. B
3	CSE	Dr. A
4	ECE	Dr. C
5	IT	Dr. B
6	CSE	Dr. A
7	ECE	Dr. C
8	IT	Dr. B
9	CSE	Dr. A
10	ECE	Dr. C

Answer:

Student Details:-

StudentID	Dept
1	CSE
2	IT
3	CSE
4	ECE
5	IT
6	CSE
7	ECE
8	IT
9	CSE
10	ECE

Dept Details:-

Dept	HOD
CSE	Dr. A
IT	Dr. B
ECE	Dr. C